

→

Digital System
Information saved in the form of numbers.

→

Calculated the converted numbers according to mentioned base.

$$(i) (15)_8 \rightarrow ()_2 = 1101$$

$$(ii) (13)_{10} \rightarrow ()_2 = 1101$$

$$(iii) (695)_{10} \rightarrow ()_3 = 221202$$

$$(iv) (244)_5 \rightarrow ()_{16} = 4A$$

$$(15)_8 \rightarrow ()_{10}$$

$$\therefore 1 \times 8^1 + 5 \times 8^0 = (13)_{10}$$

$$(13)_{10} \rightarrow ()_2$$

2	13	
2	6	1
2	3	0
2	1	1
	0	1

$$\therefore (13)_{10} \rightarrow (1101)_2$$

$$\therefore (695)_{10} \rightarrow ()_3$$

3	695	
3	231	2
3	77	0
3	25	2
3	8	1
3	2	2
	0	2

$$\therefore (695)_{10} \rightarrow (221202)_3$$

$$(244)_5 \rightarrow ()_{10}$$

$$\therefore 2 \times 5^2 + 4 \times 5^1 + 4 \times 5^0 = 50 + 20 + 4 = (74)_{10}$$

$$\therefore (74)_{10} \rightarrow ()_{16}$$

$\Rightarrow$

16	74	10 $\rightarrow$ A
16	4	4
	0	

$$\Rightarrow (74)_{10} = (244)_5 = (4A)_{16}$$

Q. Convert the following in binary

$$\rightarrow (645)_{10} \rightarrow (1010000101)_2 \quad \text{Ans.}$$

$$\rightarrow (333)_{10} \rightarrow (101001101)_2$$

$$\rightarrow (265)_{10} \rightarrow (100001001)_2$$

2	645	
2	322	1
2	161	0
2	80	1
2	40	0
2	20	0
2	10	0
2	5	0
2	2	1
2	1	0
2	0	1

2	333	1
2	166	0
2	83	1
2	41	1
2	20	0
2	10	0
2	5	1
2	2	0
2	1	1
	0	

$$(333)_{10} \rightarrow (101001101)_2$$

$$(645)_{10} \rightarrow (1010000101)_2$$

2	265	1
2	132	0
2	66	0
2	33	1
2	16	0
2	8	0
2	4	0
2	2	0
2	1	1
	0	

$$\therefore (265)_{10} \rightarrow (100001001)_2$$

→ Convert the following number in octal base

(i) $(868)_{10} = (1544)_8$

(ii) $(735)_{10} = (1337)_8$

(iii) $(297)_{10} = (451)_8$

8	868	4
8	108	4
8	13	5
8	1	1
	0	

$$(868)_{10} \rightarrow (1544)_8$$

8	735	7
8	91	3
8	11	3
8	1	1
	0	

$$(735)_{10} \rightarrow (1337)_8$$

"Happiness is when what you think, what you say, and what you do are in harmony." — Mahatma Gandhi

8	297	1
8	37	5
8	4	4
	0	

$$(297)_{10} \rightarrow (451)_8$$

→ Convert the following numbers in hexadecimal.

(i) $(255)_{10} \rightarrow (FF)_{16}$

(ii) $(178)_{10} \rightarrow (B2)_{16}$

(iii) $(97)_{10} \rightarrow (61)_{16}$

(iv) $(49)_{10} \rightarrow (31)_{16}$

16	255	15 → F
16	15	15 → F
	0	

$$(255)_{10} \rightarrow (FF)_{16}$$

16	178	2
16	11	11 → B
	0	

$$(178)_{10} \rightarrow (B2)_{16}$$

16	97	1
16	6	6
	0	

$$(97)_{10} \rightarrow (61)_{16}$$

16	49	1
16	3	3
	0	

$$(49)_{10} \rightarrow (31)_{16}$$

→ Binary Algebra

Q.

$$\begin{array}{r} 111 \\ 1011 \\ + 0101 \\ \hline 10000 \end{array}$$

As; $(1011)_2 \rightarrow (11)_{10}$

$(101)_2 \rightarrow (5)_{10}$

$\therefore 11 + 5 = (16)_{10} \rightarrow (10000)_2$ Ans.

Q.

$$\begin{array}{r} 110001011 \\ + 111000 \\ \hline 111000011 \end{array}$$

As, $(110001011)_2 \rightarrow (451)_{10}$

$(111000)_2 \rightarrow (56)_{10}$

$\therefore 451 + 56 = (507)_{10} \rightarrow (111000011)_2$ Ans.

Q.

$(23.25)_{10} \rightarrow (2)_2$

∴

2	23	1
2	11	1
2	5	1
2	2	0
2	1	1
	0	

$(23)_{10} \rightarrow (11101)_2$

2	0.25	0
2	0.50	1
	0	

$\therefore (23.25)_{10} \rightarrow (11101.01)_2$

→

$$(69.173)_{10} \rightarrow (?)_2$$

2	69	1
2	34	0
2	17	1
2	8	0
2	4	0
2	2	0
2	1	1
	0	

$$(69)_{10} \rightarrow (1000101)_2$$

2	.173	0
2	.346	0
2	.692	1
	.384	

$$\therefore (69.173)_{10} \rightarrow (1000101.001...)_2 \quad \text{Ans.}$$

→

$$(335.23)_{10} = (?)_5$$

5	335	0
5	67	2
5	13	3
5	2	2
	0	

$$(335)_{10} \rightarrow (2320)_5$$

5	0.23	1
5	.15	0
5	.75	3
5	.75	3

$$\therefore (335.23)_{10} \rightarrow (2320.1033)_5 \quad \text{Ans.}$$

→

$$(277.173)_{10} \rightarrow (?)_{14}$$

14	277	13
14	19	5
14	1	1
	0	

$$(277)_{10} \rightarrow (15B)_{14}$$

14	.173	2
14	.422	5
	.908	

$$\therefore (277.173)_{10} \rightarrow (15B.25)_{14} \quad \text{Ans.}$$

→

$$(1011011)_2 \rightarrow (28)_{10}$$

As,

$$(8)_{10} \rightarrow (1000)_2$$

Hence we will make pairs of 3

$$\begin{array}{ccccccc} 0 & 0 & 1 & 0 & 1 & 1 & 0 & 1 & 1 \\ \hline & & 1 & & 3 & & 3 & & \end{array}$$

Hence;

$$(1011011)_2 \rightarrow (133)_4$$

→

$$(1011011)_2 \rightarrow ()_{16}$$

$$\text{As, } (16)_{10} \rightarrow (10000)_2$$

Hence we will make pairs of 4

$$\begin{array}{ccccccc} 0 & 0 & 1 & 0 & 1 & 1 & 0 & 1 & 1 \\ \hline & & 5 & & 1 & & 3 & & \end{array}$$

Hence;

$$(1011011)_2 \rightarrow (513)_{16}$$

11 → 13
in a
hexadecimal
system

■

Complement of a bit

Let us assume we have a number 'N' in base 'r' having 'n' digits.

$$\begin{aligned} \text{Then } (r-1)'s \text{ of } N & \text{ is defined as;} \\ & = (r^n - 1) - N \rightarrow \text{Diminished radix complement} \end{aligned}$$

Ex

$$(67)_{10}$$

$$9's \text{ complement of } (67)_{10} = (10^2 - 1) - 67 = 32 \text{ Ans.}$$

→

Radix's Complement

The 'r's complement of an N digit number in base r is defined as;

$$= (r^n - N)$$

$$N + S = r^n - 1.$$

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Q. Calculate the ten's complement of 012398

$$\begin{aligned} \therefore 10\text{'s of } 012398 &= 10^6 - 012398 \\ &= 987602 \end{aligned}$$

999999

012398

987601 +1
987602

Q. Calculate r' and $r-1$'s complement of $(347)_8$.

7 7 7

- 3 4 7

+1 (4 3 0 $\rightarrow$ 7's complement
4 3 1 $\rightarrow$ 8's complement

Q. Calculate r' and $r-1$'s complement of $(234)_5$.

4 4 4

- 2 3 4

+1 (2 1 0 $\rightarrow$ 4's complement
2 1 1 $\rightarrow$ 5's complement

Let us assume we have to perform a subtraction of two numbers each having small n digits

The number from which we have to subtract is M and the number to be subtracted is N and the base is r .

base = r

$M - N$

n digits

$\rightarrow$ Add the r 's complement of subtrahend N to M .

$$(M - N + r^n)$$

If $M > N$

$$(M - N) + 10^n$$

discard the extra carry.

If $N > M$

$$= 10^n - (N - M)$$

$$= 10^n - (10^n - (N - M))$$

$$= N - M$$

Ex

$$23 - 12$$

10's complement of 12

∴

$$(23) + (88)$$

↳

$$\underline{111}$$

→ discard extra carry

$$\textcircled{11}$$

Ans.

$$\begin{array}{r} 99 \\ -12 \\ \hline 87 \end{array} \xrightarrow{+} 88$$

Ex

$$12 - 23$$

$$\cancel{88} - (\cancel{23} - 12)$$

Taking 10's complement again

$$12 + 77 = 89 = -11 \text{ Ans.}$$

↳ 10's complement of 23

$$\begin{array}{r} 99 \\ -23 \\ \hline 76 \end{array} \xrightarrow{+} 77$$

Q

$$(90009951)_{10}$$

Find 10's complement.

$$99999999$$

$$-90009951$$

$$\underline{99900048}$$

+1 → 99900049 → 10's complement

$$\begin{array}{r} 99 \\ -89 \\ \hline 10 \end{array} \xrightarrow{+} 11$$

Q.

$$5250 - 1321$$

As, $M > N$;

Adding 1321's 10's complement to 5250

$$5250 + 8679$$

$$\underline{13929} \rightarrow \text{discard extra carry}$$

$$3929 \text{ Ans.}$$

$$\begin{array}{r} 9999 \\ +1321 \\ \hline 8678 \end{array}$$

Q.

$$20 - 100$$

As

$$M < N;$$

Adding 10's complement of 100 to 20

$$\therefore 20 + 900$$

$$920$$

Now, taking 10's complement of 920

$$\therefore 80$$

Adding a negative sign;

$$-80$$

Ans

$$\begin{array}{r} 299 \\ - 100 \\ \hline 899 \\ + 1 \rightarrow 900 \end{array}$$

$$\begin{array}{r} 999 \\ - 920 \\ \hline 079 \\ + 1 \rightarrow 80 \end{array}$$

 $\Rightarrow$ Signed bits:

When arithmetic operations are implemented in a computer it is more convenient to use a different system for representing negative numbers.

It is referred as signed complement system.

In this system, a negative number is indicated by its complement.

Whereas the signed magnitude system negates a number by changing its sign. The signed complement system negates a number by taking its complement.

0 $\rightarrow$ +ve

$$+6 \rightarrow 00110$$

$$-13 \rightarrow 11101$$

$$-11 \rightarrow 11011$$

 $\rightarrow$

Binary Codes;

1.

Binary Coded Decimal

BCD of 46

$$\begin{array}{c} 46 \\ \swarrow \quad \searrow \\ 0100 \quad 0110 \end{array}$$

$$\begin{array}{c} 993 \\ \swarrow \quad \searrow \\ 1001 \quad 0011 \end{array} \rightarrow 100110010011$$

"There are no great things, only small things with great love. Happy are those." - Mother Teresa.

2. Excess - 3

Addition of 3 to BCD

	BCD	Excess-3
0	0000	0011
1	0001	0100
2	0010	0101
3	0011	0110
4	0100	0111
5	0101	1000
6	0110	1001
7	0111	1010
8	1000	1011
9	1001	1100

Q. BCD of $(475)_8$ is $(475)_8 \rightarrow 4 \times 8^2 + 7 \times 8 + 5 \times 8^0$
 $\rightarrow 317$

$$\therefore (475)_8 = (317)_{10}$$

BCD of 317 is

$$\therefore (317)_{10} = (475)_8 = (1100010111)$$

BCD

and excess 3

Q. BCD of $(355)_7$ is

$$(355)_7 \rightarrow 3 \times 7^2 + 5 \times 7 + 5$$

$$\rightarrow 187$$

$$\therefore (355)_7 = (187)_{10}$$

BCD of 187 is

1	8	7	BCD :	11000111
↓	↓	↓		
0001	1000	0111		
Excess-3 :	1011	1010	Excess-3 :	1110111010

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→

8 4 - 2 - 1 code

∴ 3 in 8 4 - 2 - 1

0 → 0000

$$\hookrightarrow 3 = 0(8) + 1(4) + 0(-2) + 1(-1)$$

1 → 0111

2 → 0110

$$= 4 - 1 = 3$$

3 → 0101

4 → 0100

5 → 1011

→

2 4 2 1 code

0 → 0000

1 → 0001

2 → 0010

3 → 0011

4 → 0100

5 → 0101

→

5 0 4 3 2 1 0 code

	5	0	4	3	2	1	0
0 →	0	1	0	0	0	0	1

1 →	0	1	0	0	0	1	0
-----	---	---	---	---	---	---	---

2 →	0	1	0	0	1	0	0
-----	---	---	---	---	---	---	---

3 →	0	1	0	1	0	0	0
-----	---	---	---	---	---	---	---

4 →	0	1	1	0	0	0	0
-----	---	---	---	---	---	---	---

* 5 →	1	0	0	0	0	0	1
-------	---	---	---	---	---	---	---

6 →	1	0	0	0	0	1	0
-----	---	---	---	---	---	---	---

→

Error Detection Code

Binary information can be transmitted from one device to another through wires it could be wireless.

During the transmission the bits could be damaged or lost.

To find whether received bits are correct we use error detection codes using parity bits.

Even parity

odd parity

1 0 0 0 0

1

0

1 0 0 0 1

0

1

2 0 0 1 0

1 1 1 0 0

1

3 0 0 1 1

0 1 1 0 1

0

4 0 1 0 0

1 0 1 0 0

1

5 0 1 0 1

0 0 1 0 1

0

6 0 1 1 0

1 1 0 1 0

0

7 0 1 1 1

0

1

8 1 0 0 0

0 1 0 0 0

1

9 1 0 0 1

0 0 0 1 0

0

H

Gray Code

In gray code, when we go from one number to next number only one bit is changed

0 → 0000

1 → 0001

2 → 0011

3 → 0010

4 → 0110

5 → 0111

6 → 0101

7 → 0100

8 → 1100

9 → 1101

When we convert in gray code only the first bit changes from the left and converted to the other three...

→ ASCII Character Code

$b_7 b_6 b_5 b_4 b_3 b_2 b_1$

a → 01100001 → 61

→ Register

Register is a group of binary cells.

$2^3 \text{ bits} \rightarrow 1 \text{ byte}$

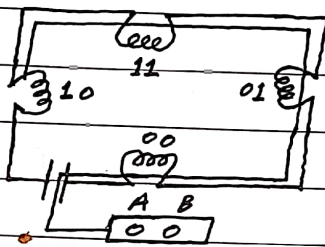
$2^{13} \text{ bits} \rightarrow 1024 \text{ byte} \rightarrow 1 \text{ kb}$

$2^{23} \text{ bits} \rightarrow 1024 \text{ kb} \rightarrow 1 \text{ Mb}$

Q.

You have given a condition that you have a room with 4 walls all having four lights.

Next, you have to switch on the light whenever any person enters in the room and switch off the light when the person exits and no. of lights will be equal to the number of persons, and the family has only four persons to enter in the room there are two switches whenever any person enters the room he touches the switch according to their code assigned.



Switch A	Switch B	$B_3 B_2 B_1 B_0$
0	0	0 0 0 1
0	1	0 0 1 0
1	1	0 1 0 0
1	0	1 0 0 0

$$\therefore B_0 = A'B'$$

$$B_1 = A'B$$

$$B_2 = AB$$

$$B_3 = AB'$$

Ans.

Q. Minimize the following boolean expression using K-Map.

$$F(A, B, C, D, E) = \sum m(0, 1, 2, 4, 6, 8, 9, 10, 12) + \sum d(16, 17, 18, 20)$$

AB \ CDE	000	001	011	010	110	111	101	100
00	0	1		2	6	7	5	4
01	8	9	11	10	14	15	13	12
11								
10	X 16	X 17	19	X 18	22	23	21	X 20

In group cell (0, 1, 8, 9) : $\bar{A}\bar{C}\bar{D}$

" " " (0, 4, 8, 12) : $\bar{A}\bar{D}E$

" " " (0, 2, 8, 10) : $\bar{A}\bar{C}E$

" " " (0, 2, 4, 6) : $\bar{A}\bar{B}E$

∴ $F(A, B, C, D, E) = \bar{A}\bar{C}\bar{D} + \bar{A}\bar{D}E + \bar{A}\bar{C}E + \bar{A}\bar{B}E$

Q. Given a function

$$F(A, B, C, D) = \sum m(1, 3, 5, 7, 9, 11, 13, 15)$$

Minimize the expression using K-Map.

Also draw the circuit with minimum number of gate.

∴

AB \ CD	00	01	11	10
00	0	1 1	1 3	2
01	4	1 5	1 7	6
11	12	1 13	1 15	14
10	8	1 9	1 11	10

↓ D

∴ $F(A, B, C, D) = D$

(2)

Circuit :

A.

B.

C.

D.

Y (output)



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Q.

$$F(A, B, C) = (A' + B)(B' + C')(A + B + C')$$

$$= (A' + B + CC')(B' + C' + AA')(A + B + C')$$

$$= (A' + B + C)(A' + B + C')(A + B' + C')(A' + B' + C')(A + B + C')$$

$$\therefore F(A, B, C) = \sum M(1, 3, 4, 5, 7)$$

		BC			
		B+C	B+C'	B'+C	B'+C'
		00	01	11	10
A	0	0	1	1	2
A'	1	1	1	1	6

→ A+B

→ C'

$$A' + B + C \rightarrow 100 \rightarrow 4$$

$$A' + B + C' \rightarrow 101 \rightarrow 5$$

$$A + B' + C' \rightarrow 011 \rightarrow 3$$

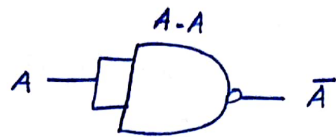
$$A' + B' + C' \rightarrow 111 \rightarrow 7$$

$$A + B + C' \rightarrow 001 \rightarrow 1$$

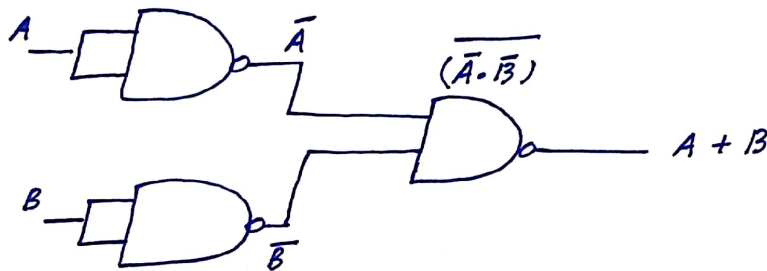
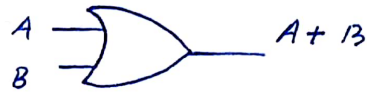
$$\therefore F(A, B, C) = (A+B) \cdot C'$$

Ans.

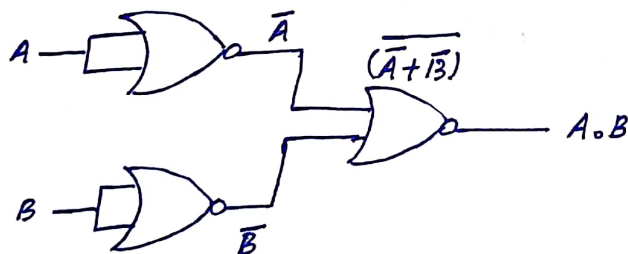
Implementing NOT from NAND



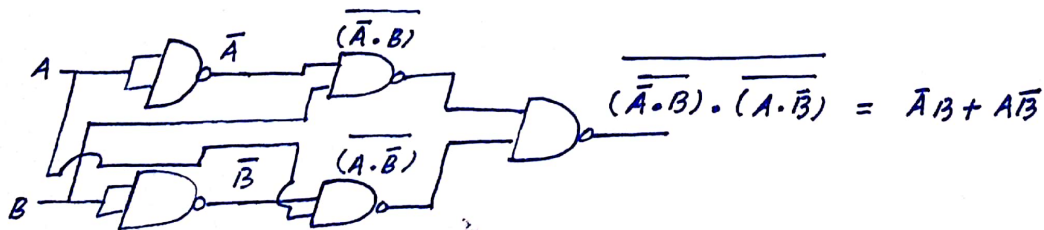
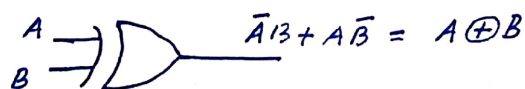
Implementing OR from NAND



Implementing AND from NOR



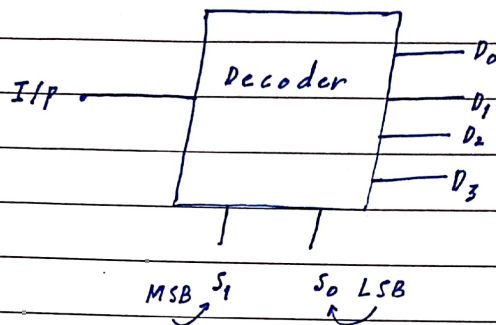
Implementing XOR from NAND



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De-multiplexers

Decoders are circuit that receive information on a single line and transmit this information on one of 2^n output lines, where n is the input control. These are called selection lines.

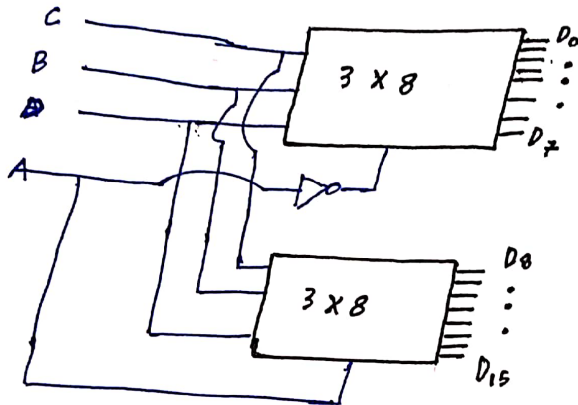


S_1	S_0	D_3	D_2	D_1	D_0
0	0	0	0	0	1
0	1	0	0	1	0
1	0	0	1	0	0
1	1	1	0	0	0

→ Implement 4×16 decoder using 2 3×8 decoders

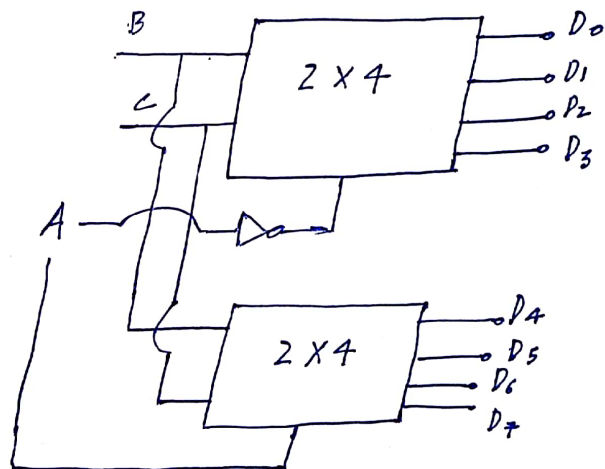
	A	B	C	D
0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1
10	1	0	1	0
11	1	0	1	1

12	1	1	0	0
13	1	1	0	1
14	1	1	1	0
15	1	1	1	1



Implement 3x8 decoder using 2x4 decoder

	A	B	C
0	0	0	0
1	0	0	1
2	0	1	0
3	0	1	1
4	1	0	0
5	1	0	1
6	1	1	0
7	1	1	1



Ans.

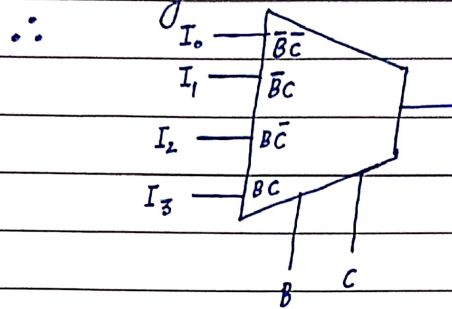
Multiplexers

Q.

Implement the boolean function

$$F(A, B, C) = \sum (1, 5, 7) \text{ using multiplexer}$$

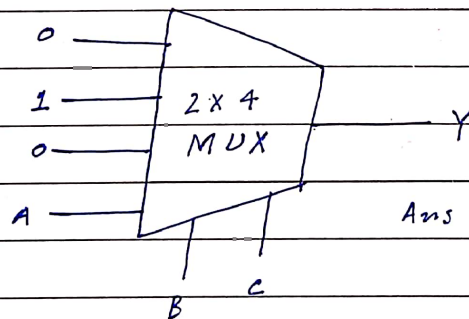
Selecting BC as select lines



$\therefore$

	I_0	I_1	I_2	I_3
	$\overline{B}\overline{C}$	$\overline{B}C$	$B\overline{C}$	BC
$\overline{A}$	0 $\overline{A}\overline{B}\overline{C}$	1 $\overline{A}\overline{B}C$	2 $\overline{A}B\overline{C}$	3 $\overline{A}BC$
A	4 $A\overline{B}\overline{C}$	5 $A\overline{B}C$	6 $AB\overline{C}$	7 ABC
	0	1	0	A

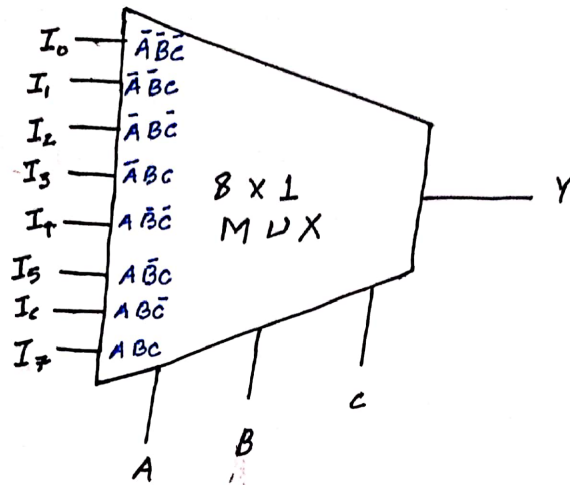
$\therefore$



Q.

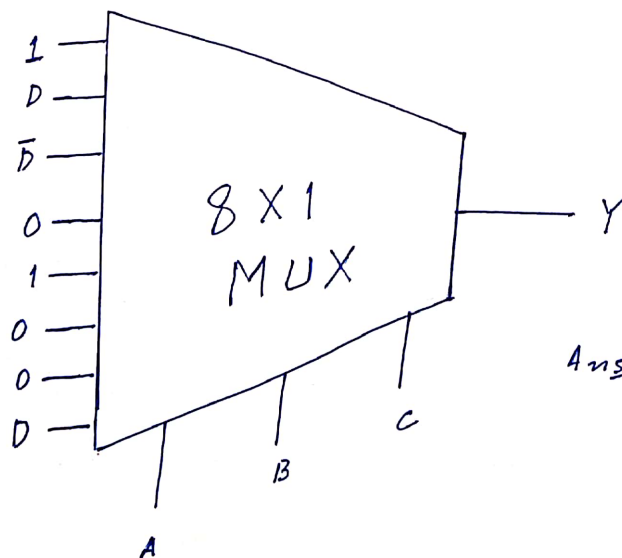
Implement following boolean function

$$F(A, B, C, D) = \sum m(0, 1, 3, 4, 8, 9, 15) \text{ using } 8 \times 1 \text{ multiplexer.}$$



Using
A B C as select
Lines.

	I_0 $\bar{A}\bar{B}\bar{C}$	I_1 $\bar{A}\bar{B}C$	I_2 $\bar{A}B\bar{C}$	I_3 $\bar{A}BC$	I_4 $A\bar{B}\bar{C}$	I_5 $A\bar{B}C$	I_6 $AB\bar{C}$	I_7 ABC
$\bar{D}$	0 $\bar{A}\bar{B}\bar{C}\bar{D}$	2 $\bar{A}\bar{B}C\bar{D}$	4 $\bar{A}B\bar{C}\bar{D}$	6 $\bar{A}BC\bar{D}$	8 $A\bar{B}\bar{C}\bar{D}$	10 $A\bar{B}C\bar{D}$	12 $AB\bar{C}\bar{D}$	14 $ABC\bar{D}$
D	1 $\bar{A}\bar{B}\bar{C}D$	3 $\bar{A}\bar{B}CD$	5 $\bar{A}BCD$	7 $\bar{A}BCD$	9 $A\bar{B}\bar{C}D$	11 $A\bar{B}CD$	13 $AB\bar{C}D$	15 $ABCD$
	1	D	$\bar{D}$	0	1	0	0	D



Ans.

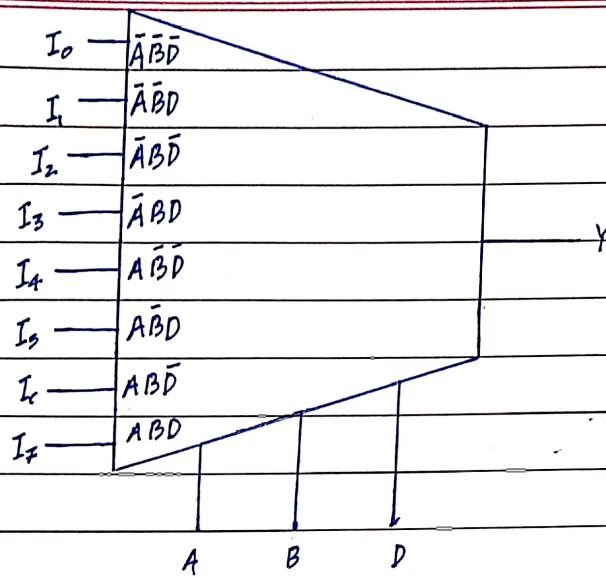
Q.

Implement the following boolean function with 8X1 multiplexer with C as input.

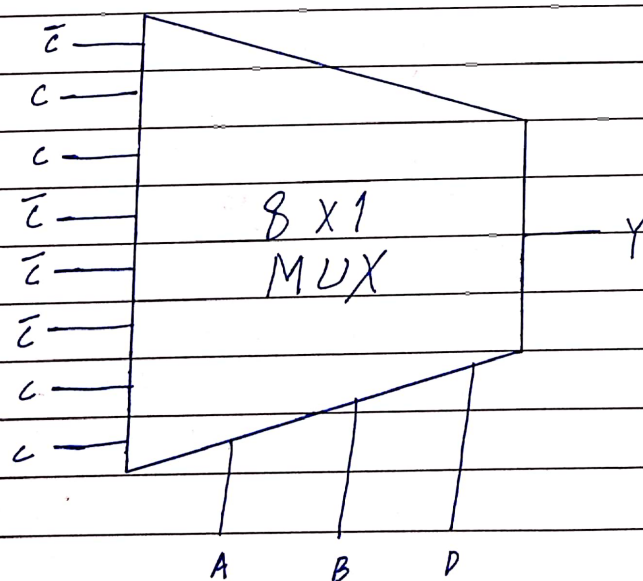


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$$F(A, B, C, D) = \sum (0, 3, 5, 6, 8, 9, 14, 15)$$

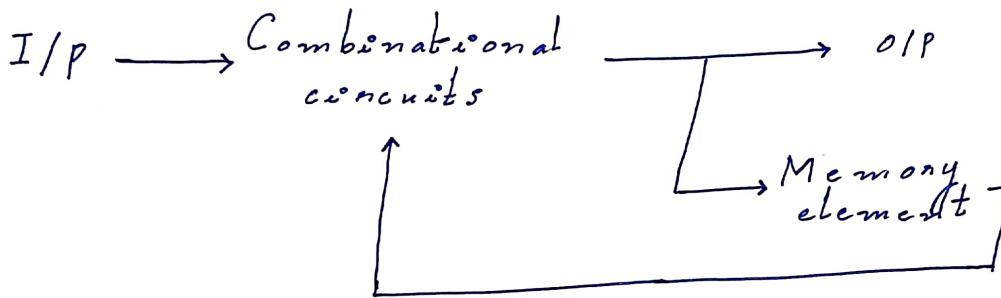


	I_0	I_1	I_2	I_3	I_4	I_5	I_6	I_7
	$\bar{A}\bar{B}\bar{D}$	$\bar{A}\bar{B}D$	$\bar{A}B\bar{D}$	$\bar{A}BD$	$A\bar{B}\bar{D}$	$A\bar{B}D$	$AB\bar{D}$	ABD
$\bar{C}$	(0)	1	4	(5)	(8)	(9)	12	13
	$\bar{A}\bar{B}\bar{C}\bar{D}$	$\bar{A}\bar{B}\bar{C}D$	$\bar{A}B\bar{C}\bar{D}$	$\bar{A}B\bar{C}D$	$A\bar{B}\bar{C}\bar{D}$	$A\bar{B}\bar{C}D$	$AB\bar{C}\bar{D}$	$AB\bar{C}D$
C	2	(3)	(6)	7	10	11	(14)	(15)
	$\bar{A}\bar{B}C\bar{D}$	$\bar{A}B\bar{C}D$	$\bar{A}BC\bar{D}$	$\bar{A}BCD$	$A\bar{B}C\bar{D}$	$A\bar{B}CD$	$ABC\bar{D}$	$ABCD$
	$\bar{C}$	C	C	$\bar{C}$	$\bar{C}$	$\bar{C}$	C	C



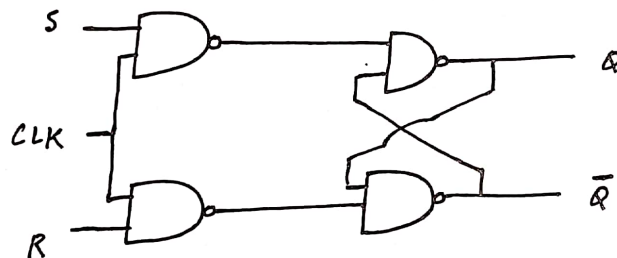
→ Sequential Circuits

Sequential Circuits are the circuits that generate output from the combinational circuits and store that output in a memory element which is further used for generating next output.



All sequential circuits use clocks.

→ S-R Flip Flop



Truth Table

S	R	Q_{n+1}	$\bar{Q}_{n+1}$
0	0	Q_n	$\bar{Q}_n \rightarrow \text{Hold}$
0	1	0	1 $\rightarrow \text{Reset}$
1	0	1	0 $\rightarrow \text{Set}$
1	1	X	X $\rightarrow \text{Invalid}$

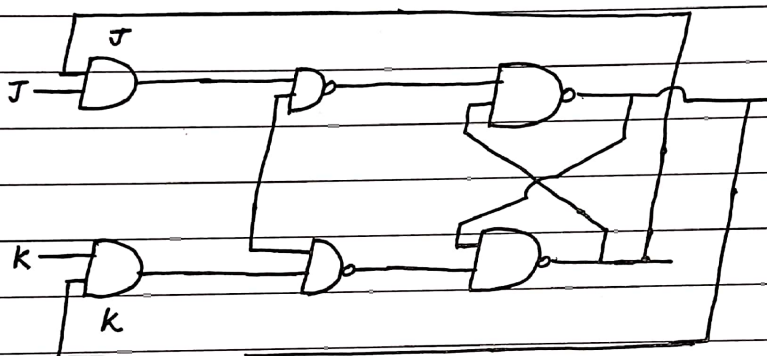
Characteristic Equation
 $Q_{n+1} = S + \bar{R}Q_n$

Excitation Table

Q_n	Q_{n+1}	S	R
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0

#

J K Flip-flop



Truth Table

J	K	Q_{n+1}
0	0	Q_n
0	1	0
1	0	1
1	1	$\bar{Q}_n$

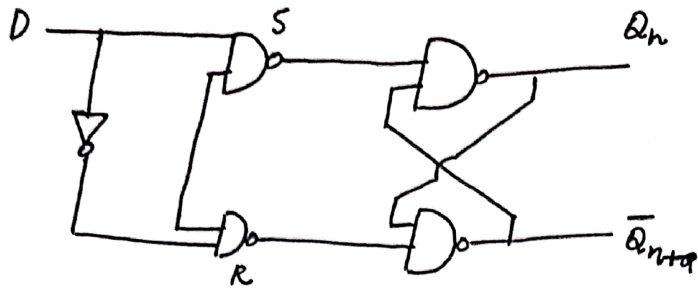
Characteristic Equation

$$Q_{n+1} = \bar{K}Q_n + J\bar{Q}_n$$

Excitation Table

Q_n	Q_{n+1}	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

→ D - Flip flop



Truth Table

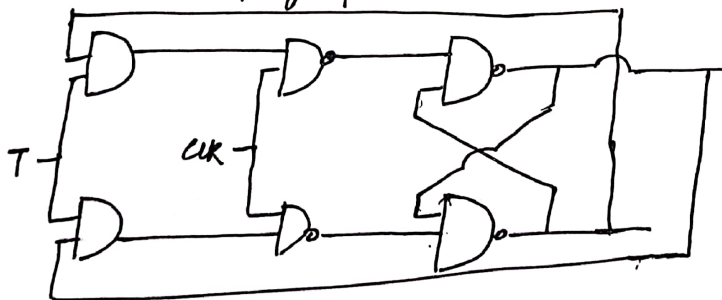
D	Q_{n+1}
0	0
1	1

Characteristic equations
 $Q_{n+1} = D$

Excitation Table

Q_n	Q_{n+1}	D
0	0	0
0	1	1
1	0	0
1	1	1

→ T - Flip-flop



Truth Table

T	Q_{n+1}
0	Q_n
1	$\bar{Q}_n$

Characteristic equation

$$Q_{n+1} = T \oplus Q_n$$



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Excitation Table

Q_n	Q_{n+1}	T
0	0	0
0	1	1
1	0	1
1	1	0

Triggering of Flip-flops

The state of a flip is switched by a momentary change in the input signals, this momentary change is called trigger, and state change is called triggering of flip flop

State Table

Implementing the state table using T flip flops

Present State		Next State		T_A	T_B
A	B	A	B		
0	0	0	1	0	1
0	1	1	0	1	1
1	0	1	1	0	1
1	1	0	0	1	1

$$\Rightarrow T_B = 1$$

$\therefore$ For T_A

A \ B	0	1
0	0 ₀	1 ₁
1	0 ₂	1 ₃

$$\Rightarrow T_A = B$$

Circuit diagram

